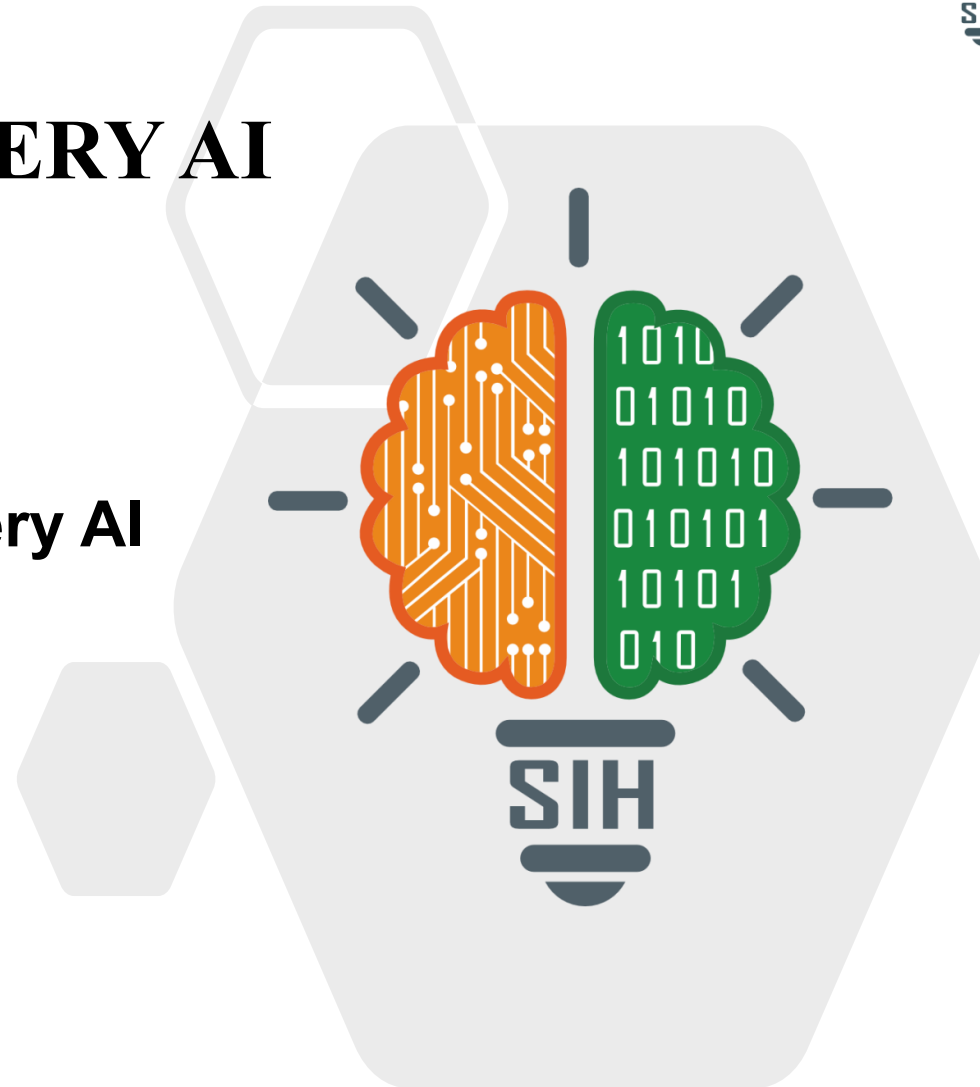


SMART INDIA HACKATHON 2026



SATQUERY AI

- Problem Statement ID – SIH26167
- Problem Statement Title – SatQuery AI
- Theme – Space Technology
- PS Category – Software
- Team ID –
- Team Name (Registered on portal) – Lunarix



Lunaris

SatQuery AI

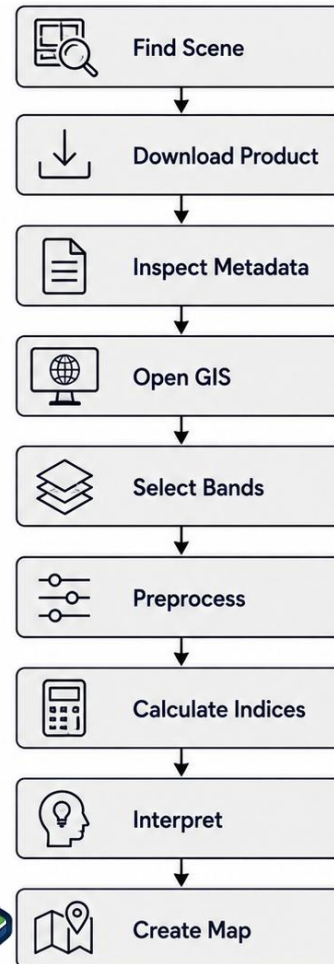
Problem

- EO data is powerful but difficult to query
- Analysis requires specialist tools and workflows
- Insights remain inaccessible to non-experts

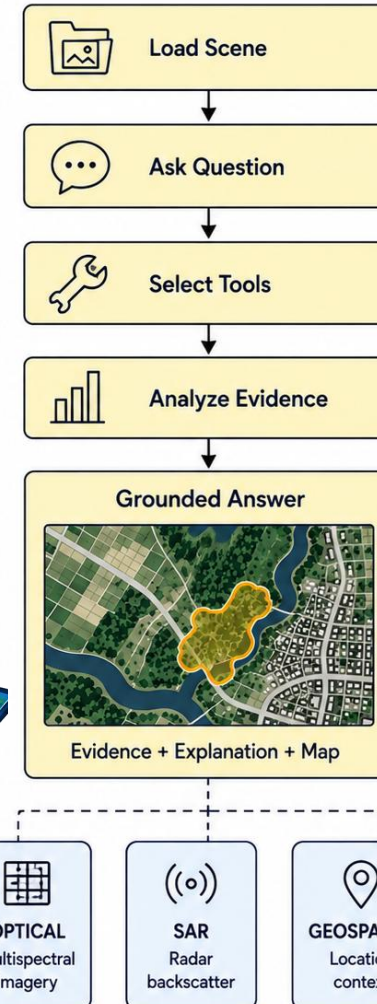
Solution

- Ask questions in natural language
- Convert intent into EO analysis
- Deliver clear, evidence-grounded insights

TRADITIONAL WORKFLOW



SATQUERY AI



USER INTENT
BECOMES
THE INTERFACE

What Makes It Different:

- Tool-driven analysis
- Multimodal Earth Observation reasoning
- Evidence-grounded outputs
- Context-aware workflows
- Reproducible analysis
- Domain-specific intelligence

KEY IDEA

User intent becomes the interface.

HOW IT WORKS

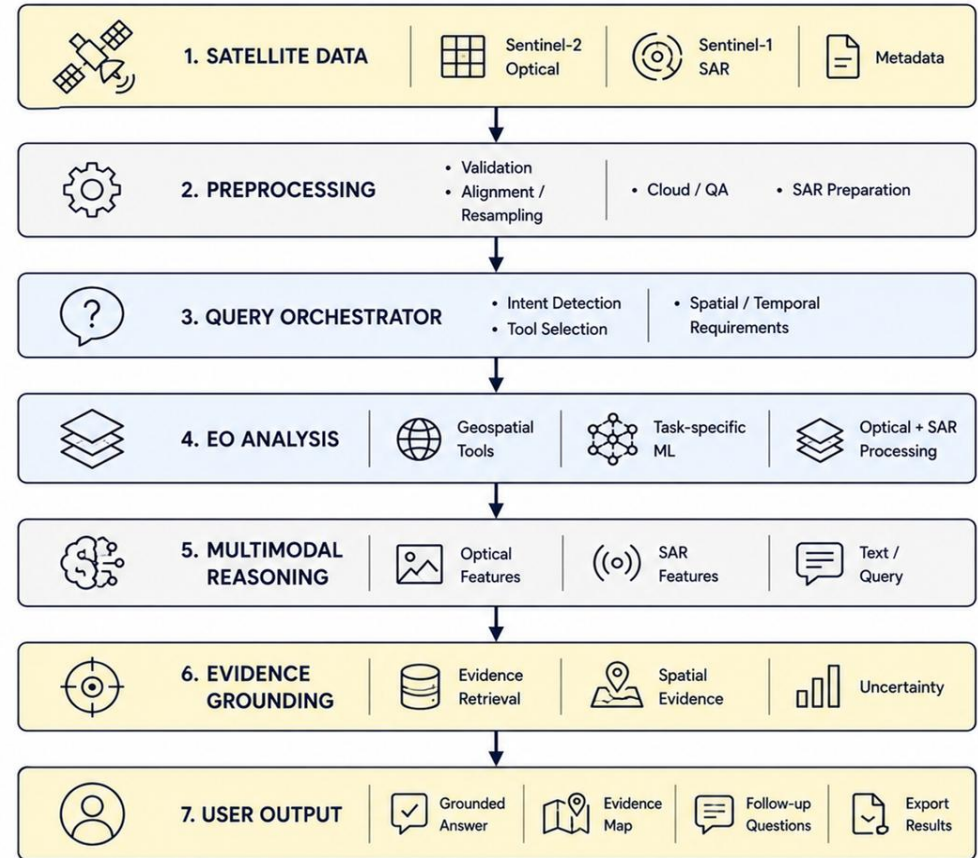
- Understand - Parse user intent
- Orchestrate - Select relevant tools
- Analyze - Process EO data
- Ground - Link claims to evidence
- Respond - Deliver result + map

System characteristics

- Modular tool orchestration
- Local / lightweight deployment pathway
- Reproducible analytical operations
- Evidence-aware output generation

Design principles:

- Separate ingestion, orchestration, analysis and reasoning into modular stages.
- Preserve modality-specific processing rather than treating all imagery as the same input.
- Keep deterministic geospatial operations alongside ML-based components.
- Pass spatial evidence through the pipeline before generating the final response.



TECHNOLOGY STACK

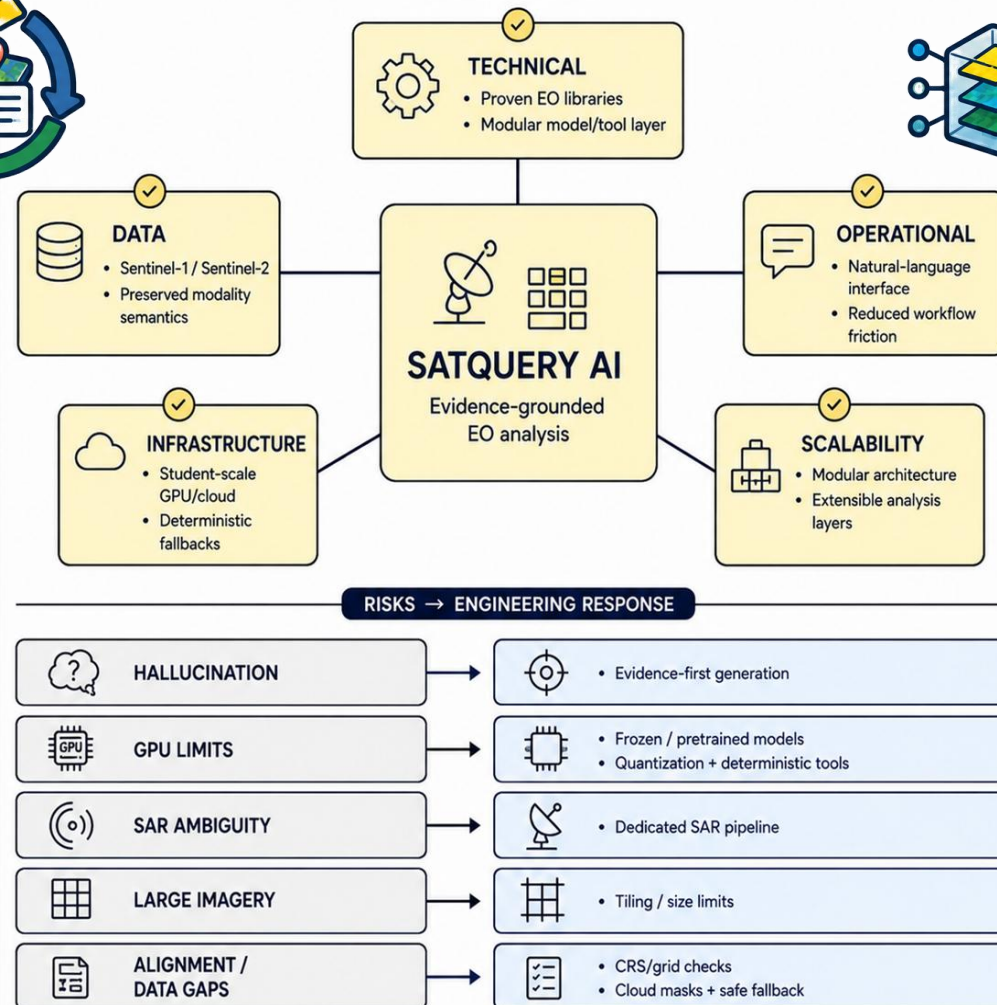


Why it is buildable:

- Mature open-source EO and geospatial processing ecosystems provide the core building blocks.
- Sentinel-1 and Sentinel-2 provide accessible multimodal data sources.
- Pretrained models can reduce the need for expensive model training.
- A modular pipeline allows capabilities to be added incrementally.



CAN IT BE BUILT?



Key Engineering Considerations:

- Evidence-first responses
- Pretrained, frozen models for limited compute
- Tiled processing for large rasters
- CRS, grid and quality validation
- Modality-specific processing for SAR uncertainty
- Lightweight, reproducible processing pipelines



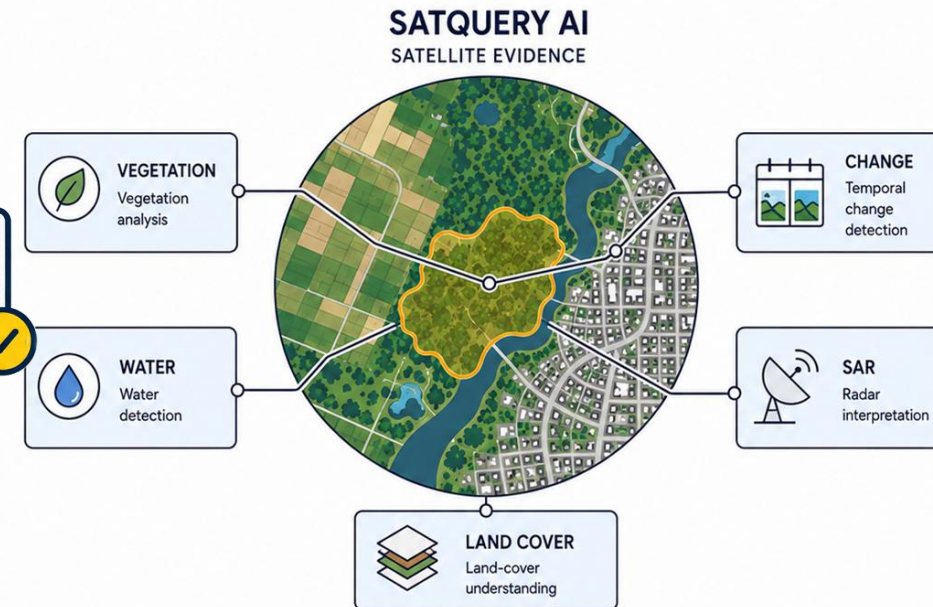
For analysts

- Faster access to relevant evidence
- Less dependence on specialized software workflows
- Easier exploration of complex EO datasets



For operational users

- Rapid access to spatial information
- More understandable analytical outputs
- Support for repeatable assessment workflows



SATELLITE EVIDENCE → ACCESSIBLE INTELLIGENCE → DECISION SUPPORT

For researchers

- Natural-language access to multimodal EO analysis
- Transparent connection between query and evidence
- A foundation for testing new EO reasoning workflows



Potential applications

- Agriculture & vegetation monitoring
- Water and environmental assessment
- Disaster response
- Land-cover analysis
- Urban change monitoring



1

Copernicus Sentinel-1 — SAR Mission

Synthetic Aperture Radar (SAR) data and mission information
<https://sentinels.copernicus.eu/web/sentinel/missions/sentinel-1>

2

Copernicus Sentinel-2 — Multispectral Mission

Multispectral optical data and mission information
<https://sentinels.copernicus.eu/web/sentinel/missions/sentinel-2>

3

BigEarthNet — Remote-Sensing Benchmark

Large-scale benchmark for remote sensing image classification
<https://bigearth.net/>

4

BigEarthNet — Dataset

Dataset and data access for BigEarthNet
<https://huggingface.co/datasets/BIFOLD/BigEarthNetv2-0/>

5

VRSBench — Vision-Language Remote Sensing Benchmark

Benchmark for vision-language models in remote sensing
<https://arxiv.org/abs/2406.19838>

6

GDAL — Geospatial Data Abstraction Library

Geospatial data translation and processing tools
<https://gdal.org/>

7

Rasterio — Raster Data Processing

Python library for raster data reading and writing
<https://rasterio.readthedocs.io/>

